

Part A:-

Qno 1 : Take a Number N and Print Numbers From N down to 1 .

Pseudocode:

DECLARE N : INTEGER

INPUT N

WHILE ~~INPUT~~ $N \geq 1$ DO

PRINT N

$N = N - 1$

ENDWHILE.

Algorithm:-

1. Start the program

2. Prompt the user to enter a number N

3. Check if it is greater than or equal to 1 then,

• print the Number

• subtract 1 from Number

• Repeat the process until Number becomes less than 1 .

5. End the program.

Qno2. Input a year and determine whether it is a leap year (Rule: divisible by 4 AND (Not divisible by 100 OR divisible by 400)).

Pseudocode.

DECLARE Year : INTEGER

INPUT Year

IF Year % 4 = 0 AND Year % 100 ≠ 0 OR
Year % 400 = 0

THEN

PRINT ("It is a leap year")

ELSE

PRINT ("It is not a leap year")

ENDIF

Algorithm.

1. Start
2. IF Input Year
3. IF Year MOD 4 = 0 AND Year MOD 100 ≠ 0 OR Year MOD 400 = 0
then print output "It is a leap year"
else,
print "It is not a leap year".
4. End.

Qno3: Take a Number, Count how many digits it has and state whether its a single digit, double digit or triple digit (or more) number

Pseudocode:-

```
DECLARE N, digits : INTEGER
```

```
DECLARE Number : STRING
```

```
INPUT N
```

```
Number = NUM TO STRING (N)
```

```
digits = LENGTH (Number)
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```
Print "The number has" digits "digits".
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You Print the multiplication tables from 1 to 10, but skip the table of any number that is a multiple of 3.

Pseudocode

DECLARE i, j : INTEGER

$i = 1$

WHILE $i \leq 10$ DO

IF $i \% 3 \neq 0$

THEN

~~multiply~~ $j = 1$

WHILE $j \leq 10$ DO

$multiply = i * j$

Print(multiply)

$j = j + 1$

ENDWHILE

ENDIF $i = i + 1$

ENDWHILE

Qno 5. Take a number, repeatedly sum its digits until the result is a single digit, and display each immediate result. For example, if the number is 123, the sum of its digits is 6.

Pseudocode:

DECLARE N, Number, Sum : INTEGER

INPUT N

WHILE $N \geq 10$ DO

Sum = 0

WHILE $N > 0$ DO

~~$N = N \text{ DIV } 10$~~

~~Sum = Sum +~~

Number = $N \% 10$

Sum = Number + Sum

$N = N \text{ DIV } 10$

ENDWHILE

Print(Sum)

ENDWHILE.